Why Startups Fail

Sector : Finance / Venture Capital

Team Information

Team ID: Section C · Group 17

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GITHUB:https://github.com/riyaseema80/SectionC_G17_WhyStartupsFail/tree/dev#-project-overview

TABLEAU DASHBOARD: Dashboard

The Investment Problem

- ## The Context

  Billions are invested in startups every year 4 yet many fail
  Investors lack a data-driven way to identify high-risk startups early

- ## The Gap

  No quantitative framework exists to flag high-failure-probability startups before capital is deployed.

- ## The Objective

  Build a screening framework to:

  - Flag high-risk startups before investing
  - Identify portfolio companies that need intervention

Startups Analysed

Crunchbase VC dataset, 1990-2014

Funding Gap

Median difference between closed vs operating firms

Failure Rate

Overall closure rate across 25 years

From Raw Data to Analysis-Ready Dataset

Source Data

Crunchbase VC dataset from Kaggle.
54,294 rows · 37 columns ·
1990-2014.

ETL Pipeline

Python processing with pandas + numpy.
Deduplicated startups, removed outliers and engineered features.

Clean Output

33,613 clean rows · 47 columns.
Exported to 10 files in data/processed/.
Notebooks 01 & 02 committed on GitHub

Data Dictionary Highlights :

status

TARGET VARIABLE

Startup outcome (operating/closed/acquired/ipo)

funding_total_usd

PRIMARY PREDICTOR

Total capital raised (USD)

funding_rounds

COUNT PREDICTOR

Number of funding rounds

is_closed (engineered)

REGRESSION TARGET

Binary: 1=failed, 0=survived

funding_tier (engineered)

SEGMENTATION

Micro/Seed/Growth/Late Stage

Key Performance Indicators

Core metrics computed from 33,613 startups across 25 years of Crunchbase data

Overall Failure Rate

closed / total × 100

5.27%

Baseline - how bad is the problem?

Funding Gap

Median operating - closed

$900K

How much underfunding predicts failure

Median Funding - Closed

Direct computation

$850K

The danger zone threshold

Median Funding - Operating

Direct computation

$1.75M

The survival baseline

Avg Rounds

mean(rounds) grouped by status

1.45 vs 1.76

Round count as investor confidence signal

Series A Survival Lift

Pre-Series A vs post-Series A

5.21% → 4.71%

Critical survival milestone

All KPIs computed in notebooks/04_statistical_analysis.ipynb · exported via notebooks/05_final_load_prep.ipynb

5 Early Signals That Predict Startup Failure

1. Single-round startups = highest risk

Most likely to fail early

HIGHEST RISK

2. Underfunding is the biggest gap

$850K vs $1.75M (closed vs operating)

$900K GAP

3. Sector risk varies sharply

Media/Gaming j 2× risk vs Enterprise Software

2× HIGHER RISK

4. Timing impacts survival

2005-2009 cohort shows peak failures

PEAK FAILURE WINDOW

5. Series A = survival milestone

Risk drops after crossing Series A

SAFER AFTER SERIES A

Failure is predictable early 4 if you track the right signals

DATA SCIENCE | STATISTICAL VALIDATION

Not Just Funding Amount - Funding Journey Determines Survival

Testing Methodology - Advanced statistical tests confirm that funding patterns - not just capital amounts - drive failure outcomes. All analyses conducted using scipy.stats and statsmodels libraries.

Mann-Whitney U Test

Closed startups received significantly less funding ($p < 0.05$)
→ Underfunding increases failure risk

Chi-Square Test

Sector and failure are dependent ($p < 0.01$)
→ Some sectors are structurally riskier

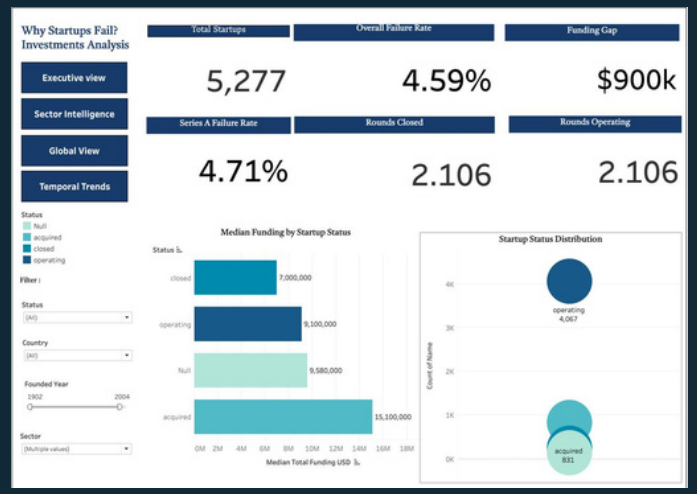
Logistic Regression

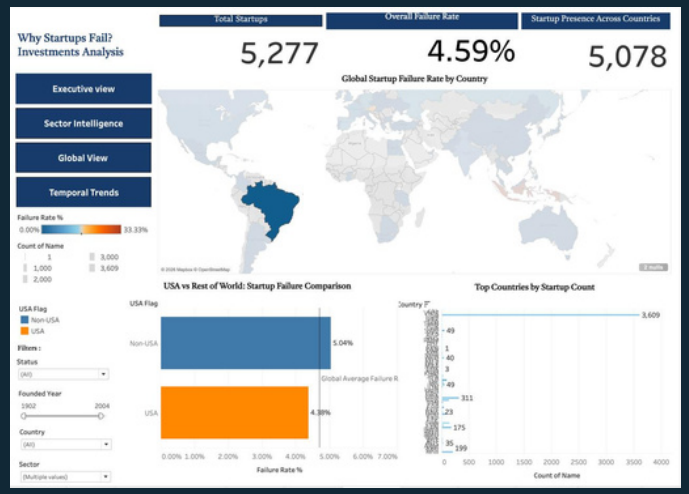
Funding low → failure risk
Funding rounds (low) failure risk (stronger effect)

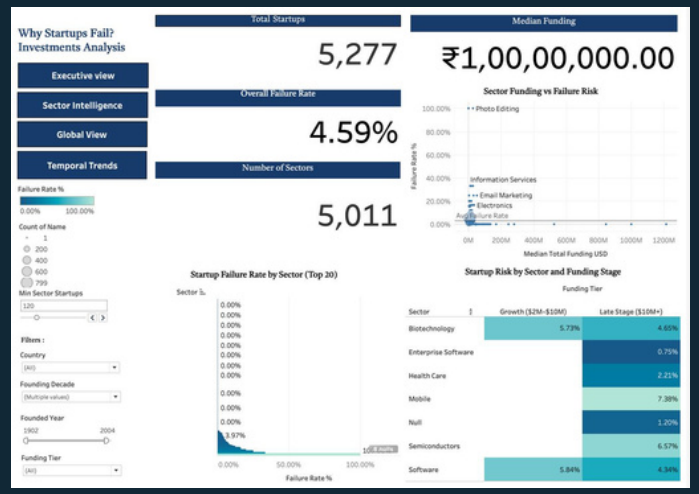
Key Finding: $p < 0.01$ statistical significance for sector as failure predictor. Funding round count provides independent predictive power beyond total capital raised.

From Raw Data to Decision-Ready Dashboard

Interactive Tableau dashboard enables real-time screening of portfolio companies against historical failure benchmarks



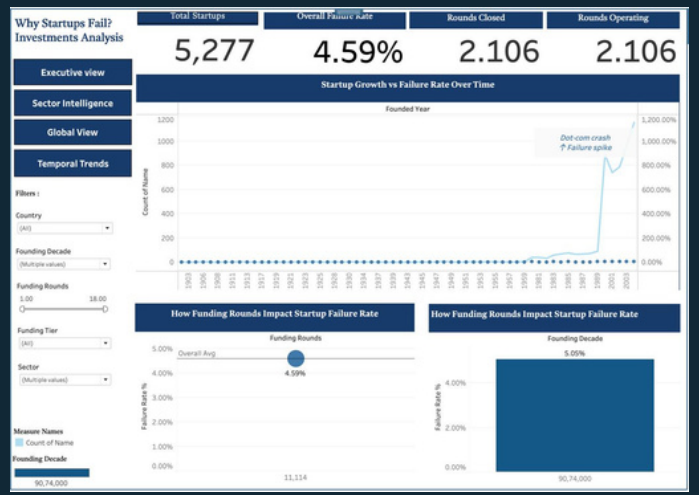




Interactive Filters : Country · Sector · Decade · Status · Funding Rounds

Tableau Public URL : Dashboard

- Failure risk is highly concentrated , not widespread across all startups
- Single funding round startups emerge as the highest-risk group
- Underfunding strongly correlates with failure across sectors
- Crossing Series A acts as a survival milestone
- More funding rounds = higher survival probability



5 Actions VC Funds Can Take Today

Implementation framework to reduce portfolio failure rate using existing investment data

Single-round startups fail most

Set a minimum round-count threshold; review single-round companies within 90 days

Expected Impact: Reduce portfolio failure rate by 10315%

Underfunding predicts closure

Require co-investor commitment for deals below the $500K seed floor

Expected Impact: Fewer underfunded deals enter the portfolio

Series A is the survival inflection

Prioritize pre-Series A mentorship, bridge capital, and introductions

Expected Impact: Higher Series A conversion; lower early-stage closure

Sector failure rates vary widely

Use sector-adjusted return hurdles; require higher multiples for Media/Gaming

Expected Impact: Better risk-adjusted portfolio returns

Macro conditions amplify failures

Track cohort failure quarterly; slow deal pace during contracting cycles

Expected Impact: Lower vintage-year concentration risk

The Cost of Not Acting

Every unflagged single-round startup is a preventable loss

Avoidable Loss

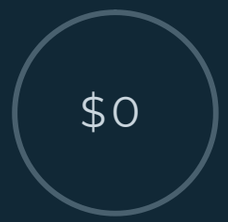
Per underfunded failed startup

Failure Reduction

Portfolio impact from round-count screening

Projected Rate

Conservative post-implementation estimate



Implementation Cost

No new data required

Financial Impact

A fund deploying $50M across 100 startups currently expects 5 failures at baseline. A round-count screening rule alone could prevent 132 of those 4 saving $1M → $2M in deployed capital per fund cycle. The screening framework requires no new data: every signal identified in this analysis is available at the time of investment decision.

Cost of Inaction

Without a data-backed screening layer, fund managers will continue to rely on pattern-matching and gut instinct - systematically over-investing in single-round, undercapitalised companies that historical data shows

What We Know, What We Don't and Where This Goes Next

Limitations

Data ends at 2014

Post-2014startup dynamics (SaaS, crypto, AI) not represented

5.27% failure rate may be understated

Survivorship bias present in old data

Causation vs correlation

Results show association, not root cause

Next Steps

01

Predictive model

TrainRandomForest/XGBoost classifier for failure probability scores

02

Extend dataset

Add post-2014 Crunchbase/PitchBook data for AI/SaaS era validation

03

Real-time dashboard

Connect Tableau to live pipeline for continuous monitoring